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Industry 5.0: Mapping the Lens from Know How to Realization

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Abstract

Industry 5.0 is considered as the latest revolution from the industrial perspective which is aiming to collaborate machine efficiency with the expertise of humans and the expected outcome will be the systems having optimized productivity and better utilization of resources. This chapter primarily addresses the basic principles of Industry 5.0, the role of its enabling technologies, operator generations up to Operator 5.0, and the evolution of Education 5.0. Furthermore, the role of Industry 5.0 and its technologies in sustainability has been explained with the help of available literature. This chapter will help in the successful integration and adaptation of enabling technologies within the existing framework of industries.

It also helps the academicians, managers, decision-makers, and researchers in the field of industrial revolutions to increase the understanding and to realize interrelationships between industrial revolutions, operator generations, and education revolutions.

Keywords: Digital technologies, IoT, industry 4.0, industry 5.0, sustainability, education 5.0, operator 5.0

2.1 Introduction

The drive toward greater globalization, shorter product life cycles, an increasing desire aimed at supportable products as well as a heightened awareness of environmental and social impacts that have prompted businesses in adopting the latest technical solutions to remain competitive [1]. Numerous organizations across the globe persist in employing conventional quality-improving approaches such as Total Productive Maintenance, Lean Six Sigma, and similar methods. Nevertheless, these methodologies prove inadequate in efficiently tackling issues regarding atmospheric emission reduction as well as product customization on a large scale [2]. In the past two decades, there has been a growing trend toward Industry 4.0 to enhance system responsiveness, minimize errors, and ensure the production of high-quality items [3]. Industry 4.0 embodies an innovative technological paradigm that integrates advancements, for example, machine learning, artificial intelligence, big data, and cybersecurity [4]. While Industry 4.0 plays a role in improving organizational efficiency by decreasing faults, reduction in waste as well as reducing labor hours, it lacks effectiveness in addressing concerns regarding environmental and social sustainability [5]. Diverse continuous improvement strategies, including Kaizen, are essential for achieving holistic enhancements in conjunction with the incorporation of related digital technologies. The 4th Industrial Revolution places substantial emphasis on technology and often ignores the man-made issues inside the entire production functionary [6, 7]. It overlooks the considerations for human cognition, emotions, and well-being, essential elements for fostering a workplace that is both friendly and accident-free. Incorporating human aspects into the production system creates a more reliable and safer work

environment, thereby contributing to the improvement of organizational social sustainability [5, 8].

The emergence of Industry 5.0 is traced back to the evolution of various revolutions. The 1st Industrial Revolution was characterized by the discovery of the steam engine, marking the beginning of manufacturing mechanization [9]. The 2nd Industrial Revolution was predominantly associated with combustion engines and electricity, placing a 19th-century focus on electrifying the mechanical production system [10]. Electricity emerged as a more convenient energy source for production compared to steam, as it eliminated the need for water and facilitated a more straightforward electricity transfer. The 3rd Industrial Revolution was centered around digitization, automation as well as the Internet. The Internet assumed roles in calculations and data transfer, marking a shift toward the digitalization of data [11]. Progress in computer technologies and electronics throughout the 20th century significantly enhanced the life of humans, introducing greater efficiency as well as regulation to various businesses. Complex industrial tasks became significantly more manageable during this period. The era also witnessed the introduction of humanoid robots, resembling humans, collaborating to achieve automated production with heightened precision. Industry 4.0 concentrated on various automation technologies such as the Internet of Things, artificial intelligence, robotics, and blockchain [4, 12]. Progress in communication and information technology has facilitated the digitization of businesses, establishing comprehensive networks that encompass CPS systems. The 5th Industrial Revolution aims to fulfill customer needs by providing personalized and customized products, catering to a diverse customer base through mass personalization. In contrast to Industry 4.0, which is centered on machines, Industry 5.0 introduces a human-centered system. This approach involves harnessing human cognitive thinking for specific tasks while delegating routine tasks to rapidly responsive and precise information technology-assisted integrated machinery as well as robots. Industry 5.0 incorporates various kaizen measures to integrate human capabilities, enhance cognitive thinking, and promote improved workplace safety [13]. In its design solution, Industry 5.0 gives top priority to people, emphasizing optimal human collaboration alongside collaborative robots (cobots). These cobots work in tandem with humans to facilitate au-

tomation in production specifically for everyone with the help of enterprise social networks. It fosters smooth collaboration between individuals and machines. Unlike programmable machines, cobots have the capability to sense and understand the presence of a person in their vicinity. In an ideal scenario, cobots are assigned to repetitive tasks and those demanding substantial physical effort, while humans handle customization and tasks requiring critical thinking [14, 15]. In its initial phases, Industry 5.0 requires a thorough understanding of factors contributing to its success, potential barriers, and the relevant technologies for successful implementation across diverse industrial sectors. Therefore, there is an urgent necessity for undertaking a comprehensive review of the 5th Industrial Revolution, providing systematic guidelines for practitioners, managers, academics, and researchers. The heightened focus of Industry 5.0 on social and environmental sustainability emphasizes the significance of conducting a systematic study of Industry 5.0 literature.

2.2 Basic Principles of Industry 5.0

Industry 5.0 signifies the latest innovation in industrial transformation, extending the groundwork established by the 4th Industrial Revolution. The fundamental theory behind Industry 5.0 centers on a more advanced human-centric, collaborative as well as sustainable model of industrial production. It places emphasis on the following core principles:

2.2.1 Human-Machine Collaboration

In variation to the 4th Industrial Revolution, which frequently prioritized automation as well as machine-centric procedures, the 5th Revolution underscores the importance of tight association between machines and humans. Its aim is to enhance human capabilities by harnessing advanced technologies rather than completely replacing human workers [16, 17].

2.2.2 Customization and Flexibility

The objective of the 5th Industrial Revolution was to create personalized, high-quality services and products while ensuring production efficiency. This is accomplished through adaptable, flexible production processes

that swiftly adapt to evolving customer requirements as well as market dynamics [13].

2.2.3 Sustainability and Responsible Manufacturing

Addressing mounting environmental apprehensions, Industry 5.0 advocates for sustainability and conscientious manufacturing practices. It concentrates on optimizing resource utilization, curbing waste, and minimizing the environmental impact across the supply chain. Many times, blockchain technology is employed to augment accountability as well as transparency in responsible manufacturing [18].

2.2.4 Evolving Technologies

Industry 5.0 persists in harnessing a variety of enabling advanced technologies, encompassing the Internet of Things, artificial intelligence (AI), Virtual reality, augmented reality, and advanced robotics. These technologies facilitate improved operations, decision-making, and overall efficiency [11].

2.2.5 Data-Driven Insights

In Industry 5.0, analytics and big data hold considerable importance, serving as the basis for well-informed decision-making, process optimization, and predictive maintenance. These elements contribute to enhanced efficiency and effectiveness in organizational operations [19].

2.3 Technologies and Their Roles in Industry 5.0

The 5th revolution incorporates a diverse type of advanced technologies that contribute to the transformation of production and industrial processes. These technologies collaborate to establish intelligent, interconnected, and collaborative systems. Table 2.1 depicts technologies and their area of application regarding Industry 5.0. Figure 2.1 also depicts different associated technologies of Industry 5.0.

Below are pivotal technologies and their roles in Industry 5.0 in detail:

Table 2.1 Technologies and their area of application regarding industry 5.0.

Technology	Area of application	References
Artificial intelligence	Optimizing operational efficiency, optimize resource usage, design and development of sustainable products by simulation, facilitate human-machine collaboration, environmental monitoring and management, Mass customization, decision support system	[19–22]
Internet of things	Connectivity and integration of systems, real-time monitoring, Predictive maintenance, data analytics, supply chain optimization, creation of smart products and services,	[26, 28, 29, 31, 32]
Blockchain	Supply chain traceability, assist in smart contracts for automation, decentralized data storage and security, promote interoperability and collaboration among diverse stakeholders, Intellectual property protection	[30, 36–39]
VR/AR	Training and skill development, provides real-time guidance, remote assistance and maintenance training, safety training and simulations, facilitate collaborative design and planning sessions, better customer engagement	[46–49]

Technology	Area of application	References
Robotics and cobots	Collaborate with humans without posing any safety risks, can provide flexible automation, safety features like collision detection, vision systems, Cost effective automation	[54–57]
Cloud computing	Cost-effective data storage solutions, data processing and analytics, digital twin implementations, to achieve cost efficiency,	[63–66]
Edge computing	Reduces the latency in data processing, assists in data processing in real time, provides scalability and flexibility, facilitates machine learning and on-device analytics	[72–75]
Additive manufacturing	Rapid prototyping of new designs, production of unique and tailored items, assist in manufacturing complex geometries and intricate designs,	[82–85]
Cybersecurity	Safeguards sensitive data, assists in maintaining network security, supply chain security and endpoint security, supply chain integration and collaboration, Identity and access management	[90–92]



Figure 2.1 Technologies of industry 5.0.

2.3.1 Artificial Intelligence

AI is employed in data analysis, predictive maintenance, and autonomous decision-making, elevating overall operational efficiency. AI assumes a pivotal role in shaping and optimizing various facets of Industry 5.0 [19, 20]. The major roles of artificial intelligence in Industry 5.0 are mentioned below.

Artificial intelligence assists in optimizing operational efficiency. It enables predictive maintenance, helping to anticipate equipment failures and schedule maintenance before issues arise, minimizing downtime and optimizing operational efficiency. Machine learning algorithms can analyze vast amounts of data generated by industrial processes to identify patterns and optimize workflows, leading to more efficient and streamlined operations [21]. Furthermore, AI can optimize resource usage by an-

alyzing data on energy consumption, raw material utilization, and other inputs. This optimization contributes to sustainability efforts and reduces waste in industrial processes [19]. AI can also contribute to the design and development of sustainable products by simulating various scenarios and providing insights into the environmental impact of different design choices. This can lead to the creation of products that are more resource-efficient and eco-friendly [22]. AI can also facilitate human-machine collaboration. Collaborative robots, or cobots, equipped with AI capabilities can work alongside human workers, enhancing productivity and safety in manufacturing environments. AI enables these robots to adapt to changing tasks and collaborate seamlessly with human counterparts [16, 17, 23]. AI systems can also help in enabling autonomous systems. Technologies such as machine vision and natural language processing are crucial for developing autonomous systems. In Industry 5.0, these systems can include autonomous vehicles, drones, and other smart machines that operate with minimal human intervention [19]. AI systems can also assist in environmental monitoring and management. AI-powered sensors and monitoring systems can track environmental parameters in real time. This capability is valuable for industries aiming to minimize their environmental impact by ensuring compliance with regulations and identifying opportunities for improvement [24]. AI systems can help in customization and personalization for industries. AI can facilitate mass customization by analyzing customer preferences and market trends. This allows industries to produce customized products on a large scale, meeting individualized consumer demands efficiently [25]. AI can also help decision support systems. It provides advanced analytics and decision support tools that assist management in making informed decisions. This includes forecasting demand, optimizing supply chains, and responding to dynamic market conditions [26]. AI systems can also assist in adaptation and continuous learning. AI systems in Industry 5.0 are designed to learn and adapt continuously. This adaptability is crucial for responding to changing conditions, improving performance over time, and staying resilient in dynamic industrial environments [27].

2.3.2 Internet of Things (IoT)

Various aspects of manufacturing and supply chains incorporate IoT sensors and devices, offering instantaneous data for monitoring as well as control. IoT assumes a vital role in Industry 5.0, contributing to the heightened connectivity, intelligence, and collaboration that characterize this next phase of industrial evolution [28]. The major roles of IoT in Industry 5.0 are mentioned below.

IoT helps connectivity and integration of systems. IoT devices enable seamless connectivity among various components in industrial systems. This includes sensors, actuators, machines, and other devices that can communicate and share data in real time. Integration of IoT technologies allows for a holistic view of the entire industrial ecosystem, promoting interoperability among different systems and components [29]. IoT can assist in realtime monitoring and control over various systems. It facilitates real-time monitoring of equipment, processes, and assets. Sensors embedded in machinery and industrial environments collect data on parameters such as temperature, pressure, and performance, enabling prompt identification of issues and predictive maintenance. Real-time data also supports remote monitoring and control, allowing operators to manage and optimize processes from anywhere.

Predictive maintenance tasks can be done easily by using IoT-enabled systems. IoT enables predictive maintenance by continuously monitoring the condition of equipment [30]. By analyzing data patterns and detecting anomalies, predictive maintenance systems can predict when machinery is likely to fail, allowing for timely intervention and reducing downtime [31]. Data analytics and decision support can be eased by IoT support. The vast amount of data generated by IoT devices is leveraged for advanced analytics. Machine learning algorithms can analyze this data to derive actionable insights, supporting better decision-making in areas such as production planning, resource allocation, and quality control [32]. IoT technologies can assist in supply chain optimization. IoT technologies improve visibility and traceability across the supply chain. This includes tracking the movement of raw materials, components, and finished products. Enhanced supply chain visibility leads to improved efficiency, re-

duced delays, and better responsiveness to changes in demand [33]. Furthermore, IoT enables the creation of smart products and services. Products can be embedded with sensors and communication capabilities, allowing them to gather data on their usage and performance. This data can then be used to improve product design, provide personalized services, and enhance customer experiences [34]. IoT also enables communication and collaboration from diverse entities in the industrial ecosystem, including machines, humans, and other smart devices. This interconnectivity sustains adaptive and responsive systems capable of swiftly adjusting to changing conditions [35].

To sum up, in Industry 5.0, IoT facilitates improved connectivity, data-informed decision-making, predictive capabilities, and collaboration, establishing the groundwork for more intelligent, adaptive, and interconnected industrial systems.

2.3.3 Blockchain

The utilization of blockchain technology aims to achieve supply chain transparency and traceability and ensure responsible manufacturing by generating tamper-resistant records of transactions and product origins. In Industry 5.0, blockchain plays a central role in improving transparency, security as well as efficiency across various aspects of industrial operations [36, 37]. Blockchain, a decentralized and distributed ledger technology, can bring several benefits to Industry 5.0: Firstly, blockchain can help supply chain traceability. Blockchain facilitates complete traceability throughout the supply chain by documenting each transaction or transfer of products on an unchangeable ledger. This clearness is vital to guarantee the authenticity and origin of products, addressing concerns such as counterfeiting, and adhering to regulatory requirements [38]. Secondly, blockchain can assist in smart contracts for automation. Smart contracts, which are executed by themselves along with conditions that are directly encoded into code, have the potential to automate diverse processes within Industry 5.0. They can streamline and enforce agreements, such as automated payments, eliminating the necessity for intermediaries and thereby reducing delays while enhancing efficiency [39]. Thirdly, blockchain can facilitate decentralized data storage and security.

Blockchain offers a secure and decentralized means of storing sensitive data. In Industry 5.0 execution, data integrity as well as security remain crucial, the tamper-resistant characteristics of blockchain guarantee that data remains safeguarded against unauthorized access and modifications [40]. Blockchain also has the potential to promote interoperability and collaboration among diverse stakeholders in the industrial ecosystem. By offering a shared and standardized platform for recording and accessing data, it nurtures trust and collaboration among entities that may harbor mutual distrust [41]. Utilizing blockchain also enables the establishment of a decentralized system for certifying and verifying the quality of products. This is especially pertinent in industries where maintaining high-quality standards is paramount, such as in manufacturing and healthcare [39]. In the context of decentralized energy grids, blockchain can support the transparent and secure exchange of energy between various participants, including producers and consumers. This peer-to-peer energy trading can optimize resource usage and promote sustainability [42].
Blockchain can also help in Intellectual property protection. Leveraging blockchain can safeguard intellectual property rights by establishing an unchangeable record of ownership and origin. This is particularly beneficial in industries where innovation and intellectual property serve as essential assets [43]. Industry 5.0 may see the rise of decentralized autonomous organizations, where decision-making processes are governed by smart contracts on a blockchain. This can lead to more democratized and transparent governance structures within industrial entities [44].
Blockchain has the potential to grant individuals or organizations greater control over their data. In Industry 5.0, this could give rise to novel business models where data is regarded as an asset, allowing individuals or entities to securely and transparently monetize or share their data [41].
Blockchain can be applied to track the provenance of components and materials in manufacturing processes. This ensures that manufacturers have a clear record of the origin and history of each component, aiding in quality control and compliance [45].

While blockchain technology holds great promise, it is essential to note that its adoption in Industry 5.0 is still evolving, and practical implementations will depend on various factors, including regulatory considerations and industry-specific requirements.

2.3.4 Virtual Reality and Augmented Reality

Virtual and augmented reality technologies can be employed for improving interactions among machines and humans, and for training and troubleshooting, by offering visual and interactive guidance [46]. Virtual reality and augmented reality are pivotal for the implementation of Industry 5.0, revolutionizing the design, execution, and maintenance of industrial processes. The following outlines the key roles of AR and VR in Industry 5.0.

VR and AR have a major role in the training and skill development of human resources. Virtual Reality is utilized for immersive training experiences, allowing workers to simulate and practice tasks in a virtual environment. This is particularly beneficial for training in complex or hazardous environments [47]. Augmented reality provides real-time guidance and overlays digital information onto the physical environment, aiding workers in task execution. It can offer step-by-step instructions or display relevant data directly to the controller. VR and AR also assist in maintenance and repair by remote assistance. Virtual Reality is employed for remote assistance and maintenance training. Technicians can use VR headsets to visualize equipment and receive guidance from experts located elsewhere, improving the efficiency of troubleshooting and repairs. Augmented reality can provide on-the-spot information and guidance during maintenance tasks. Technicians can view digital overlays on equipment, such as schematics or repair instructions, enhancing accuracy and speed in the maintenance process [48]. VR and AR can also assist in design and prototyping. Virtual Reality enables immersive design reviews and prototyping, enabling engineers and designers to visualize and interact with 3D models in a virtual space. This fosters collaboration and aids in identifying potential issues before the creation of physical prototypes [49]. Augmented reality facilitates the real-time visualization of design modifications within the physical environment. Designers can superimpose digital models onto real-world spaces, simplifying the assessment of how proposed changes will integrate into the existing environment. VR and AR technology can also help in remote collaboration. Virtual Reality enables remote collaboration by creating virtual meeting spaces where teams can interact and collaborate in a shared environment. This is par-

ticularly valuable for global teams or during situations where physical presence is challenging [50]. Augmented reality supports remote collaboration by overlaying digital annotations or holograms onto the real world during video calls. This allows experts to guide on-site workers or provide remote assistance. VR and AR technology can also provide great help in safety training and simulations. Virtual Reality is used for safety training simulations, immersing workers in realistic scenarios to train them on emergency procedures, hazard recognition, and safety protocols. Augmented reality provides real-time safety information and alerts in the workplace. For instance, operators can get warnings related to potential dangers or navigate safely through a complex environment [51]. Apart from that, data visualization can be easily done with the help of VR and AR technologies. Virtual Reality allows for the immersive visualization of complex datasets. Engineers and analysts can explore and understand large datasets in 3D, gaining deeper insights [52]. Augmented reality can overlay real-time data onto physical objects. For instance, maintenance technicians can view equipment health metrics or performance data directly on the machinery they are monitoring [53]. Both AR and VR also facilitate collaborative design and planning sessions. Team members can share and interact with 3D models, improving communication and understanding among stakeholders [46, 52, 53]. VR and AR can also in better customer engagement. Virtual Reality is employed for virtual product showcases and demonstrations, providing customers with immersive experiences to explore products or services. Augmented reality improves product information for customers by superimposing digital content, such as specifications or user reviews, onto physical products [51, 52].

In summary, AR and VR technologies in Industry 5.0 contribute to improved training, collaboration, visualization, and efficiency across various industrial processes, leading to enhanced productivity and innovation.

2.3.5 Robotics and Cobots

Advanced robots are employed in various manufacturing and logistics tasks, working alongside human operators to improve efficiency and productivity. Cobots that are also known as collaborative robots, play a cru-

cial role in the implementation of Industry 5.0 by promoting alliances involving machines and humans, addressing the changing requirements of contemporary industrial environments [54, 55]. The roles of cobots in Industry 5.0 are mentioned below.

Firstly, Collaborative robots are particularly designed for working alongside humans in a shared workspace. They come equipped with sensors and safety features, ensuring they can collaborate with humans without posing any safety risks [17, 56]. This collaborative nature enhances overall productivity and flexibility in manufacturing processes. Secondly, cobots can provide great help in flexible automation. Unlike traditional industrial robots that often require specialized environments and extensive programming for each task, cobots are known for their flexibility. They can be easily programmed and reprogrammed for different tasks, making them suitable for small-batch production, agile manufacturing, and environments with changing requirements [57]. Furthermore, cobots are crafted with integrated safety features like collision detection, vision systems, and torque and force sensors. These attributes assist cobots in identifying and reacting to the presence of humans or unforeseen obstacles, ensuring a secure working environment [58]. Cobots are also well-suited for tasks that involve precision, repetition, and consistency. Their utilization can be applied in numerous applications that include assembly, quality inspection, testing, and pick-and-place tasks. Their adaptability makes them suitable for industries with diverse product lines [59]. Cobots can also help reduce physical strain among workforces. Cobots are engineered to manage physically demanding or repetitive tasks, alleviating the physical strain on human workers. This can lead to improved ergonomics in the workplace and help prevent injuries associated with tasks that are better suited for automation [60]. In addition, Cobots are often designed to be easily integrated into existing production lines. Their quick deployment time allows businesses to implement automation solutions without lengthy downtimes, enabling a faster return on investment [57]. Cobots can also assist in skill enhancement and empowerment. Cobots are created to enhance human capabilities rather than replace workers. They can undertake routine or physically challenging work, which enables the human workforce to focus more on value-added and complex tasks. This not only boosts job satisfaction but also offers

chances for skill enhancement within the workforce [58]. From the perspective of Industry 5.0, which emphasizes collaboration and connectivity, cobots are well-aligned with these principles. They can be integrated into smart factories where they communicate and collaborate with other machines, sensors, and systems to optimize overall production efficiency [61]. Cost-effective automation can also be achieved with the help of cobots. Cobots frequently present a more cost-effective option compared to traditional industrial robots, expanding the accessibility of automation to a broader array of businesses, including small as well as medium-sized enterprises termed SMES. This democratization of automation contributes to the widespread adoption of Industry 5.0 principles [62]. In addition, Certain cobots come with learning capabilities, enabling them to adjust to modifying conditions along with learning from human operators. This feature aligns with the adaptive nature of Industry 5.0 systems, where continuous improvement and learning are integral [57].

In summary, cobots in Industry 5.0 contribute to a more flexible, collaborative, and efficient industrial landscape, where human workers and machines work together synergistically to optimize productivity and adapt to changing production requirements.

2.3.6 Cloud Computing

Cloud services facilitate data storage, access, and analysis, making it easier to process and share information across organizations. Cloud computing contributes an imperative and transformative role in Industry 5.0 by providing a scalable, flexible, and interconnected computing infrastructure [63]. Major aspects of the role of cloud computing in Industry 5.0 are mentioned below.

Firstly, cloud computing offers scalable and cost-effective data storage solutions [64]. During the implementation of Industry 5.0, where massive amounts of data are generated through numerous sources, cloud storage provides a centralized repository that facilitates easy and secure access to data from different locations. Secondly, cloud computing has a major role in data processing and analytics. The cloud provides powerful computing resources that can be leveraged for data processing and analytics [65].

Industry 5.0 relies on advanced analytics for extracting effective insights out of enormous datasets, to enable improved decision-making, process optimization, and predictive maintenance. Thirdly, cloud computing can help in better scalability as well as flexibility. Cloud platforms do help businesses in scaling their computing resources based on demand in the concerned area. This scalability is crucial for industries dealing with fluctuating workloads or those undergoing rapid growth or transformation, aligning with the dynamic nature of Industry 5.0 [66]. Apart from that, cloud computing facilitates seamless collaboration to provide a shared as well as accessible platform for data as well as applications. In Industry 5.0, where collaboration between different entities is essential, cloud-based tools help enabling realtime distribution of information as well as collaborative decision-making [67]. Along with the proliferation of IoT devices in the implementation phase of Industry 5.0, cloud computing works as a source for the collection, processing, as well as analysis of data through these devices. Cloud platforms help to provide the essential infrastructure to manage as well as derive insights out of the extensive amounts of data processed by using IoT sensors [64]. Cloud computing also assists in edge computing integration. Cloud computing complements edge computing in the scenario of Industry 5.0. While edge computing handles data processing nearer to the source, cloud provides centralized processing and storage, ensuring a balanced and efficient distribution of computing resources [68]. Cloud computing can also help in digital twin implementations, and cloud computing facilitates the generation and control of digital twins, which are virtual replications of physical processes or systems. In Industry 5.0, digital twins control the computational power, storing capacity, and accessibility of the cloud, allowing for real-time monitoring and simulation [69]. Cloud service providers make substantial investments in security measures, enabling businesses to securely store and process sensitive industrial data. Cloud platforms also assist in compliance with industry-specific regulations and standards, ensuring data integrity and privacy [70]. Cost efficiency can also be achieved by cloud computing technology. Cloud computing operates on a real-time payment model, enabling businesses to do payments solely for the resources that they actively utilize. This sort of cost-efficiency is particularly advantageous for Industry 5.0, where optimization of operational costs is a key consideration. Cloud computing also accelerates the deployment of ad-

vanced technologies and solutions. In Industry 5.0, where rapid innovation is essential, the cloud facilitates the quick implementation of evolving technologies, for example, machine learning, advanced analytics, and artificial intelligence [64]. The cloud also enables remote access and management of industrial processes and systems. This is crucial in Industry 5.0 scenarios where remote monitoring, control, and maintenance are essential for optimizing operations [71].

In summary, cloud computing is the backbone of all enabling technologies enforcing Industry 5.0, providing the required digital infrastructure for storage of data, processing, collaboration as well as innovation. It supports the interconnected and intelligent systems that characterize the next stage of industrial evolution.

2.3.7 Edge Computing

Edge computing decentralizes the processing of data, leading to quicker response times and alleviating the load on central data centers. It serves an essential role in the implementation of Industry 5.0 by placing computing resources in proximity to the data source, facilitating real-time processing, minimizing latency, and supporting the distributed and interconnected characteristics of contemporary industrial systems [72, 73]. Below mentioned are important attributes for the role of edge computing in Industry 5.0.

Firstly, Edge computing reduces the latency in data processing by processing information closer to the data source [74]. In Industry 5.0, where real-time responsiveness is critical, reduced latency enables faster decision-making and improved system performance. Secondly, edge computing also assists in data processing in real time. Industry 5.0 is concerned about the generation of massive data through multiple sources, which includes IoT sensors and devices. Edge computing allows for real-time processing of this data at or near the source, enabling quick analysis and immediate actions [75]. Thirdly, edge computing helps optimize the utilization of bandwidth through the processing of data locally and transmitting essential information only toward centralized cloud servers. This is especially advantageous in situations where cost considerations and band-

width limitations come into play [76]. Furthermore, enhancing security and privacy, edge computing processes sensitive data locally instead of transmitting it to centralized cloud servers. This decentralized approach minimizes the risk of data breaches, ensuring that crucial information remains within the local environment [77]. In addition, Edge computing also supports the development of autonomous systems in Industry 5.0. Devices and machines equipped with edge computing capabilities can generate local results based on real-time information, enhancing the autonomy as well as efficiency of industrial processes [78]. Edge computing also provides scalability and flexibility in handling diverse and distributed workloads. In Industry 5.0, where the scale of operations can vary, edge computing allows for adaptable and efficient resource allocation [73]. Edge computing is also resilient to network instability or interruptions. In environments where network connectivity may be unreliable, having local processing capabilities ensures that critical functions can continue even in the absence of a stable connection to the cloud [72]. Edge computing also provides help in integration with IoT devices. In Industry 5.0, edge computing is intricately linked with the IoT that facilitates continuous integration with a variety of IoT devices, which also enable live monitoring, proper control, as well as management of these devices at the network's edge [79]. In terms of cost efficiency, Edge computing has the potential to cut down on expenses related to data transfer and storage. Through processing the data locally and transmission of only pertinent data toward the cloud, businesses can optimize their utilization of cloud resources, potentially leading to reduced costs associated with data transfer [72]. Edge computing also facilitates machine learning and on-device analytics, empowering devices for analyzing and responding to data locally. This is especially valuable for applications where instantaneous insights are essential, such as predictive maintenance and anomaly detection [80]. Edge computing is also well-suited for industrial environments that may be remote or have harsh conditions. Processing data locally reduces the dependence on a centralized data center, making it more practical for scenarios with limited access to cloud infrastructure [81].

In summary, edge computing in Industry 5.0 complements cloud computing by providing localized processing capabilities, reducing latency, enhancing security, and supporting the real-time demands of modern indus-

trial systems. It constitutes a vital element of the distributed and intelligent architecture that defines Industry 5.0.

2.3.8 Additive Manufacturing (3D Printing)

3D printing also known as Additive manufacturing, is employed for swift prototyping and personalized production, aligning with Industry 5.0's emphasis on customization. It has an imperative role in transforming conventional manufacturing techniques within the context of Industry 5.0 [71, 82]. Below are essential facets of the involvement of 3D printing and additive manufacturing in Industry 5.0.

Firstly, in terms of customization and personalization, 3D printing supports the production of exceptionally tailored and personalized products. In Industry 5.0, where there is an increasing focus on meeting individualized customer needs, 3D printing enables the production of unique and tailored items [83]. Furthermore, Industry 5.0 places a premium on innovation, and 3D printing is instrumental in the rapid prototyping of new designs. This allows for quick iteration and testing of concepts, speeding up the product development cycle [84]. Additive manufacturing assists in manufacturing complex geometries and intricate designs. Additive manufacturing stands out in crafting intricate designs and complex geometries, which might pose challenges or be impossible with traditional manufacturing methods. This capability holds significant value for the creation of innovative and optimized products in Industry 5.0 [82]. Secondly, 3D printing enables on-demand manufacturing, allowing companies to produce items as needed. This reduces the need for large inventories, minimizes storage costs, and supports a more agile and responsive production model [84]. Additive manufacturing also helps in supply chain resilience as flexibility and resilience in the supply chain take center stage in Industry 5.0. Additive manufacturing serves an imperative role in enhancing the resilience of the supply chain by supporting distributed or local manufacturing, decreasing reliance on centralized production facilities, and minimizing the repercussions of disruptions [85]. Thirdly, additive manufacturing produces great help in sustainability and material efficiency. Additive manufacturing produces less waste when compared to traditional subtractive manufacturing techniques. This inherently sus-

tainable process utilizes only the necessary material for constructing the final product, aligning with Industry 5.0's emphasis on sustainability [86]. Apart from this, within Industry 5.0, 3D printing reinforces the idea of digital inventory. Instead of storing tangible goods, digital files for products can be archived, and by utilizing 3D printing, these items can be produced on-demand wherever required, facilitating distributed manufacturing models [87]. The utilization of additive manufacturing supports the development of tangible prototypes commencing digital models. Via Industry 5.0, these physical prototypes can be harnessed for testing and validation within virtual environments, thereby improving the seamless integration of physical and digital design processes [87]. Utilizing 3D printing involves creating custom tooling and jigs, enabling the swift and cost-efficient development of specialized tools to enhance diverse manufacturing processes. This contributes to heightened efficiency and adaptability in operations [85, 87]. Within Industry 5.0, the medical and healthcare sectors also extensively employ additive manufacturing. This technology facilitates the creation of implants, prosthetics, and medical devices tailored to individual patients, demonstrating its versatility and profound impact on human well-being [88]. 3D printing also supports collaborative design and co-creation processes. Teams distributed across different locations can work on the same digital models, facilitating collaboration and innovation in Industry 5.0 [87, 89]. Finally, Industry 5.0 involves continuous innovation, and 3D printing allows for experimentation with new materials. This opens the door to material innovations that can improve product performance, durability, and sustainability [87].

In summary, additive manufacturing and 3D printing are integral to Industry 5.0's pursuit of customization, agility, sustainability, and innovation. The technology enhances the manufacturing landscape by enabling new design possibilities and more adaptive production methods.

2.3.9 Cybersecurity

With increased connectivity, cybersecurity is critical to protect data, systems, and processes from potential threats. The role of cybersecurity in Industry 5.0 is paramount as digitalization and connectivity become pervasive in industrial methods. Implementation of Industry 5.0 mostly pri-

oritizes intelligent, interconnected, and collaborative systems, the imperative for robust cybersecurity measures becomes vital to assure the availability, confidentiality, and integrity of both system and data [90, 91]. Below are crucial points concerning the role of cybersecurity in Industry 5.0.

Firstly, in terms of data protection, Cybersecurity safeguards sensitive data generated and processed in Industry 5.0. This includes protecting intellectual property, customer information, and other confidential data from unauthorized access, manipulation, or theft [92]. Secondly, cybersecurity also assists in maintaining network security, supply chain security, and endpoint security. Industry 5.0 relies heavily on interconnected systems and networks. Cybersecurity measures are essential to secure these networks, prevent unauthorized access, and defend against cyber threats such as malware, ransomware, and phishing attacks [91]. As Industry 5.0 emphasizes supply chain integration and collaboration, cybersecurity becomes crucial in protecting against cyber threats that may target the interconnected ecosystem of suppliers, manufacturers, and distributors [33]. Given the widespread adoption of IoT devices and intelligent sensors in Industry 5.0, prioritizing the security of endpoints becomes crucial. Cybersecurity measures are needed to protect these devices from compromise, ensuring the overall integrity of the industrial network [91]. Secondly, cybersecurity also assists in identity and access management (IAM) as well as in incident response and management. In Industry 5.0, IAM systems have a pivotal role in overseeing and regulating access to crucial systems and data. Implementing strong authentication and authorization mechanisms helps prevent unauthorized access to sensitive information [90, 91]. Industry 5.0 environments need robust incident response and management capabilities. Cybersecurity measures should include protocols for detecting, responding to, and recovering from cybersecurity incidents, minimizing downtime and potential damage [92]. Furthermore, Security concerns must be included during the initial design of Industry 5.0 enforcing systems. This involves embedding security into the architecture, protocols, and processes to establish robust and secure systems. Cybersecurity also assists in continuous monitoring. Ongoing monitoring of network traffic, system activities, and user behavior is vital for promptly detecting and addressing cybersecurity threats in

real time. This proactive approach helps prevent and respond to potential security incidents promptly [90]. Industry 5.0 systems must adhere to cybersecurity regulations and industry standards. Compliance measures help ensure that organizations meet legal requirements, protecting them from potential legal and financial consequences [90]. Cybersecurity also provides great help in data integrity and authenticity. The implementation of cybersecurity measures is vital in Industry 5.0 to ensure the reliability as well as authenticity of the data. This involves implementing mechanisms to detect and prevent data tampering, ensuring the reliability of information used in decision-making processes [91]. As human factors continue to be a notable cybersecurity concern, It is imperative to implement awareness as well as training programs for educating the workforce on cybersecurity techniques, the dangers of social engineering attacks, as well as the significance of fostering a security-conscious culture [91]. Finally, cybersecurity assists in collaboration and information sharing also. Industry 5.0 environments benefit from collaborative efforts and information sharing within the cybersecurity community. Sharing threat intelligence and best practices assists businesses to remain aware concerning developing threats as well as enhancing their overall security posture [90].

In conclusion, cybersecurity within Industry 5.0 necessitates a thorough and adaptable approach encompassing the safeguarding of data, networks, devices, and systems against a diverse array of cyber threats. It is an integral component of building a resilient and secure foundation for the intelligent and interconnected industrial systems of the future.

Industry 5.0 strives to achieve a balance among cutting-edge technologies as well as human expertise and this endeavor aims to create more gratifying and imaginative roles for workers, streamline production processes to be both efficient and tailored, and promote sustainable manufacturing practices. This collaborative, human-centric, and data-driven approach holds the ability to reshape industries and bolster their overall resilience in the face of a rapidly evolving world.

2.4 Operator 5.0

Operator generations have been revolutionized from Operator 1.0 to Operator 5.0. Figure 2.2 enumerates operator generation evolutions. The era of Operator 1.0 is generally characterized by workers doing "dextrous and manual work" along with occasional assistance from manual-controlled machines. During the Operator 2.0 phase, humans engage in "assisted work," utilizing computer tools like NC operating systems and CAx tools. Moving into the generation of Operator 3.0, humans take part in collaboration along with robots, computer tools, and other machinery, establishing what is commonly termed robot-human collaboration. Finally, the generation of Operator 4.0 signifies the "Industry 4.0 operator," embodying a skilled and smart operator who, whenever required, conducts "work aided" through machinery. This generation introduces a novel engineering and design philosophy used for adaptive manufacturing processes, which emphasizes on automation to enhance cognitive, sensorial, and human physical capabilities over the incorporation of cyber-physical systems [93–95]. Regarding adaptive automation and human-cyber-physical systems (HCPS), Operator 4.0 demonstrates proficiency during the implementation of an automation-aided system that fosters a symbiotic relationship between humans and automation [93]. This contributes to the establishment of socially maintainable workers within the framework of a smart factory which is human-centric [94, 96].

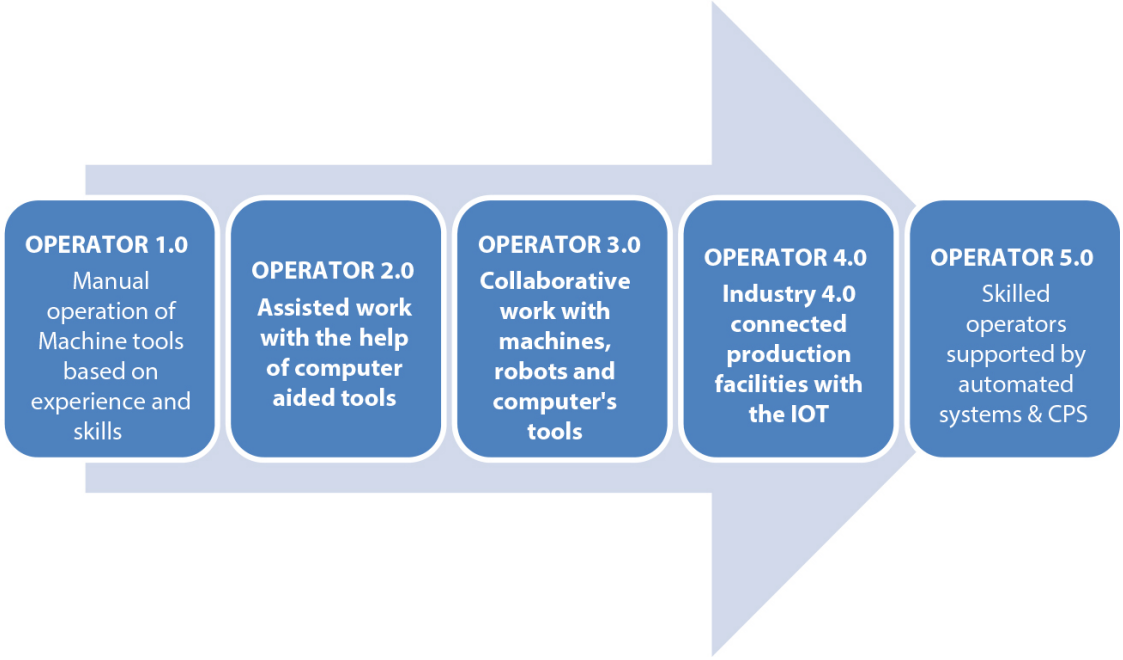


Figure 2.2 Operator generations evolutions.

Furthermore, there exists a vision for the Resilient Operator 5.0, delineated into two facets: 1st is prioritizing the cultivation of "self-resiliency" for the manpower, and 2nd is focusing on "system resiliency" throughout human-machine systems.

'Self-resiliency pertains to the occupational health and safety, encompassing biological, physical, cognitive, and psychological aspects, of every shop floor employee, along with a focus on enhancing their productivity [96]. Table 2.2 depicts the comparative analysis of operator revolutions.

This forthcoming industrial revolution means to introduce an era of "cobots," which is nothing but the succeeding generation of already existing robots that are equipped with either innate knowledge or rapid learning abilities, incorporating a human touch into robotic productions [57, 58]. Cobots, or collaborative robots, give precedence to safety and adhere to risk requirements by being aware of human presence. They can perceive, comprehend, and sense humans and understand the expectations and objectives of their operators. Like trainees, collaborative robots also observe and learn human techniques, carrying out tasks once they have acquired the necessary skills, much like human operators. Those involved in this revolution must acknowledge the irretrievable developmental trend as well as understand the crucial time window for successfully adjusting along with the amended situation [17, 61, 62].

Table 2.2 Comparative analysis of operator revolutions.

Operation revolution	Type of work	Details
Operator 1.0	Manual work	Manual operations on machine tools with the help of skills and experience
Operator 2.0	Assisted work	Use of computer-aided tools such NC, information systems
Operator 3.0	Collaborative work	Cooperative work of machines, robots, and computer tools
Operator 4.0	Connected IoT with production facilities	4th Industrial Revolution connects IoT with production facilities
Operator 5.0	Human-CPS	Skilled labors supported by automated tools, toward human-CPS.

2.5 Education 5.0

The right to education is a fundamental human entitlement, and the effective sharing of knowledge has long been a vital standard. Figure 2.3 illustrates different education revolutions. Over the last ten years, education systems have undergone notable changes due to the rise of the Internet of Things (IoT) and information communication technology (ICT). The integration of sensors and the utilization of artificial intelligence (AI) for data processing have laid the groundwork for the next stage in the development of education systems [97]. Education has historical ties to industrial revolutions, and Industry 4.0 has seen significant ef-

forts directed toward attaining Education 4.0. The objective is to enrich the learning experience via the amalgamation of the Internet of Things and Information communication technology (ICT) [98, 99].

Nevertheless, despite numerous advancements over traditional education systems, the pursuit of personalized tutoring, distinctive educational experiences, and gamebased learning has given rise to the conceptualization of Education 5.0 that represents a forward-thinking approach aimed at integrating advanced information and communication technology, i.e., ICT into the education system to elevate the experience of learning and overcome obstacles to individualized education [100, 104].

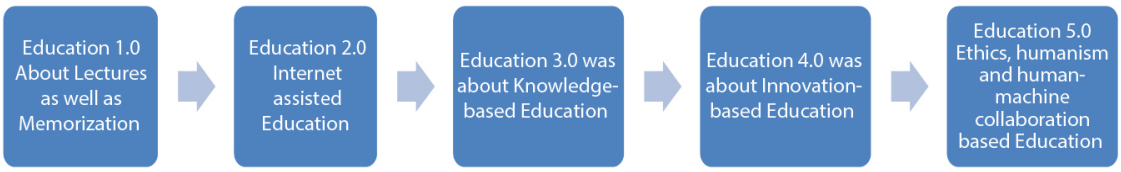


Figure 2.3 Education revolutions.

2.5.1 Historical Background of Education 5.0

2.5.1.1 Education 1.0

The first educational revolution, commonly referred to as conventional classroom instruction, was categorized by the tutor as the prime resource of knowledge. In Education 1.0, students were expected to memorize and reproduce the provided data, with rote learning being the central focus. Instructional methods in this era predominantly involved lecture-style teaching, with minimal integration of technology in the classroom. Students were assigned tasks like note-taking, memorization, and completing assessments and exams that assessed their recall skills. The teaching approach underscored the authority of the teacher, and students were required to adopt passive learning roles [102, 103].

The shortcomings observed in Education 1.0 enlighten the path of Education 2.0, which incorporates and brings technology into the teaching space. This shift aimed to move away from commencing rote learning toward collaborative learning and more interactive approaches. Education 2.0 incorporates technology for improvement in the overall ex-

perience of learning, fostering inclusivity, as well as making education approachable to a wider audience.

2.5.1.2 Education 2.0

Education 2.0 signifies an advancement in the educational scenario, addressing the flaws of Education 1.0. The introduction of technology into classrooms completely transformed the learning experience to a more collaborative and dynamic approach. These tools complement traditional teaching methods, elevating the engagement and interactivity of the learning process [104, 105]. Moreover, Education 2.0 has created opportunities for learners to keenly engage in a self-learning journey, fostering creativity as well as independent thinking through the collaborative and interactive use of technology [106, 107].

Despite the advantages of Education 2.0, it exhibited few limitations which enlighten the path to the evolution of Education 3.0. A considerably significant limitation of Education 2.0 was its tendency to view technology merely as an adjunct to conventional learning techniques. Teamwork and collaborative learning were missing in Education 2.0 also and it led to the next generation of Education, i.e., Education 3.0 [108–111].

2.5.1.3 Education 3.0

Education 3.0 contributes a significant progression in teaching, learning, and education. It included interactive activities and discussion in the classroom as the learners watched lectures as well as the completion of assignments done at home. A key focus of Education 3.0 is on promoting collaboration and teamwork, urging students to actively participate in their educational journey. This strategy supports shared learning experiences among students and the cultivation of essential skills such as critical thinking, collaboration, and communication [112].

Despite its numerous advantages compared to earlier educational models, Education 3.0 exhibited many limitations that enlightened the path of Education 4.0. The limitations include very little utilization of technological resources only for the delivery of content and it lacks emphasis on mental health issues, stress, and the overall well-being of learners.

2.5.1.4 Education 4.0

Education 4.0 harnesses advanced technologies such as blockchain, big data and analytics, cloud computing, IoT, virtual and augmented reality, artificial intelligence, and 5G networks for the betterment of the overall experience of learning and teaching experience. ML and AI assist in personalizing the teaching and learning experience after proper analysis of student's data and then doing adjustments in the pace of instruction and aligning it with the personalized needs and abilities of each student [113–116]. To provide a better and interactive learning experience, augmented and virtual reality have been utilized and they could showcase real-world cases and scenarios and help students to gain hands-on experience [117–119].

Despite bringing about significant positive changes and integrating technology, Education 4.0 encounters notable shortcomings that have prompted considerations for Education 5.0. One prominent drawback of Education 4.0 is its constraint of accessibility, because it depends significantly on technology as well as on the internet, thereby creating hurdles for learners without access to these resources. Table 2.3 enumerates a comparison of education evolutions.

Table 2.3 Comparison of education evolutions 1.0 to 5.0.

Evolution	Time period	Comparative details
Education 1.0	Pre-20th century	Passive, teacher-centered, rote learning
Education 2.0	20th century	Student-centered, critical thinking, technology integration
Education 3.0	Early 21st century	Personalized, digital literacy
Education 4.0	Ongoing	Industry collaborative teaching and interdisciplinary skills
Education 5.0	Futuristic	Holistic, emotional intelligence, lifelong learning

2.5.1.5 Education 5.0

Education 5.0 is expected to prioritize personalized and adaptive learning, with technology serving as a complement to the human experience rather than a replacement. Furthermore, Education 5.0 will incorporate the latest technological advancements, including quantum computing, blockchain, and big data. It will stress an interdisciplinary approach to learning, integrating various fields of study and fostering increased collaboration between students and teachers [120–122].

2.6 Industry 5.0 and Sustainability

The field of research in Industry 5.0 is in its early stages, making statistical predictions imprecise due to its developmental phase. Nevertheless, if the initial trends continue, sustainability within the realm of Industry 5.0 is supposed to attract a significant portion of the scientific community's attention. Consequently, approaches to sustainability must adopt a more

comprehensive perspective, wherein social considerations become as significant as economic and environmental factors [9, 18, 123].

It is crucial to acknowledge that Industry 5.0 remains in its infancy, lacking widespread recognition and a firmly established definition, with its characteristics still undergoing evolution. On the contrary, Industry 4.0 is a well-settled approach that signifies the incorporation of digital technology into production systems [6, 18, 123–125]. Nevertheless, when drawing conclusions from overarching trends and discussions in sustainability, certain potential differences among Industry 4.0 as well as the theoretical findings of Industry 5.0 become apparent, especially regarding sustainability, as encapsulated in the following research topics.

1. Social sustainability: Social sustainability could become a focal point in Industry 5.0, placing greater emphasis on issues like community engagement, diversity and inclusion, worker safety, fair wages, and labor rights.
2. Human-centric design: In Industry 5.0, there might be a focus on prioritizing the empowerment of workers and overall well-being, addressing the ethical and social effects of adopting technology.
3. Resilience: In the context of Industry 5.0, there could be efforts to initiate and implement industrial procedures focused on optimizing resource utilization, reducing waste and emissions, and adhering to the principles of the circular economy. This may involve incorporating clean and renewable energy sources, lowering greenhouse gas emissions, and mitigating the environmental impact associated with industrial activities.
4. Artificial intelligence: The capability of Industry 5.0 to notably improve the sustainability of industries lies in optimizing resource utilization, boosting operational efficiency, championing sustainable product design, facilitating environmental monitoring, and assessing social impact through the application of artificial intelligence.
5. Circular economy: In the framework of Industry 5.0, there may be backing for localized and community-based production methods, with the goal of nurturing economic resilience, empowering communities, and advancing sustainability.

6. Sustainable development: In the context of Industry 5.0, there may be a trend toward adopting a holistic approach that integrates sustainability principles across the entire value chain. This entails considering the broader societal and environmental implications as part of the commitment to sustainable development.

Although certain sustainability trends are beginning to surface in Industry 5.0, the concept is still evolving and lacks widespread recognition. The characteristics and implications of Industry 5.0 for sustainability are subjects of ongoing discussions, debates, and developments. Given the continuous evolution of the field of sustainability, it becomes imperative to examine how future industrial revolutions or paradigms can contribute to sustainable development, addressing the intricate challenges facing our society and the planet.

Considering that Industry 5.0 aims to go beyond mere prosperity and well-being, placing emphasis on sustainability and the comprehensive transformation of the industrial landscape, this undertaking poses a significant challenge. Addressing this challenge will require substantial changes, involving the development of new concepts and technologies—a trend that is evident in the noticeable rise in scientific literature and journals dedicated to this evolving domain.

2.7 Conclusion

In this chapter, the implementation challenges of Industry 5.0 have been addressed, and how various enabling technologies and concepts of Operator 5.0 and Education 5.0 are interrelated with Industry 5.0. It has been explained how operator generation evolutions have changed from manual operations to finally automated systems hybrid with human intelligence resulting in collaborative robots and how the education sector has evolved from memorization concepts in Education 1.0 to finally human-machine collaboration-based education by implementing technologies of virtual reality and augmented reality. Apart from that, some light has been focused on from a sustainability perspective also considering goals of social sustainability.

Challenges have been identified in the effective implementation of Industry 5.0. For example, the adoption of cutting-edge technologies often demands substantial investments in new hardware, software, and training for the workforce. Furthermore, integrating diverse technologies and systems from different vendors can pose challenges in terms of compatibility and seamless communication. Also, With the increased connectivity and data exchange, there are heightened risks of cybersecurity threats. Ensuring the security of sensitive industrial data is crucial. Regarding workforce adaptation, the transition to advanced technologies may necessitate reskilling and upskilling the existing workforce. Resistance to change and the requirement for a skilled workforce can pose limitations. Additionally, there are ethical and regulatory challenges, as new technologies may raise ethical concerns, and regulatory frameworks might struggle to keep pace with technological advancements, resulting in uncertainties and challenges in compliance. As a future perspective, to overcome the limitations mentioned above, more focus is required toward interoperability issues, data security, privacy concerns, training needs of manpower, failure-proof connectivity, proper address of regulatory challenges, and measurement of environmental impact along with amalgamation with green manufacturing technologies.

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Note

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